

SEG-Y_r2.1: Extended Textual Header Clarification

SEG Technical Standards Committee

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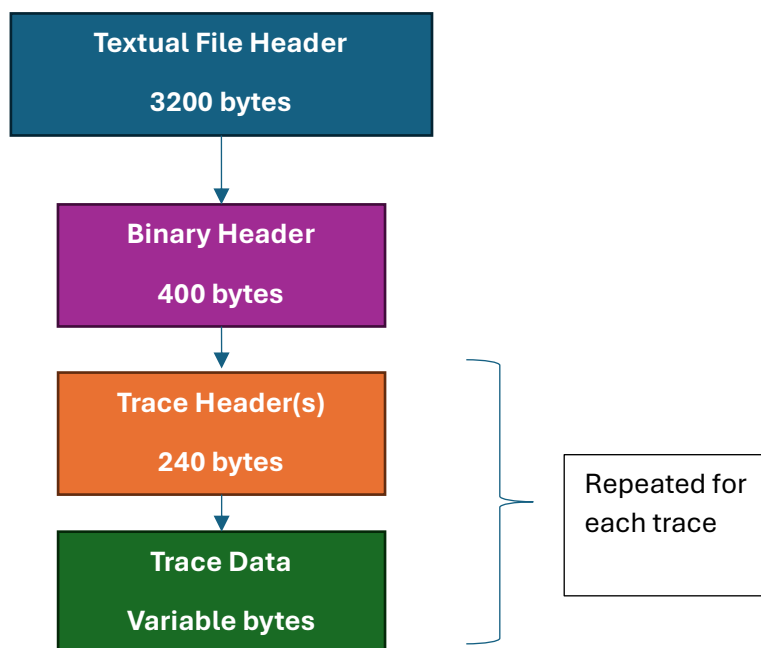
Extended Textual Header Clarification

This document is designed to clarify the implementation of Extended Textual Headers in a SEG-Y rev 2.1 format file. Extended Textual Headers allow for one or more documents to be added into the header of the SEG-Y file, with each document resulting in one or more Extended Textual Header blocks within the SEG-Y file.

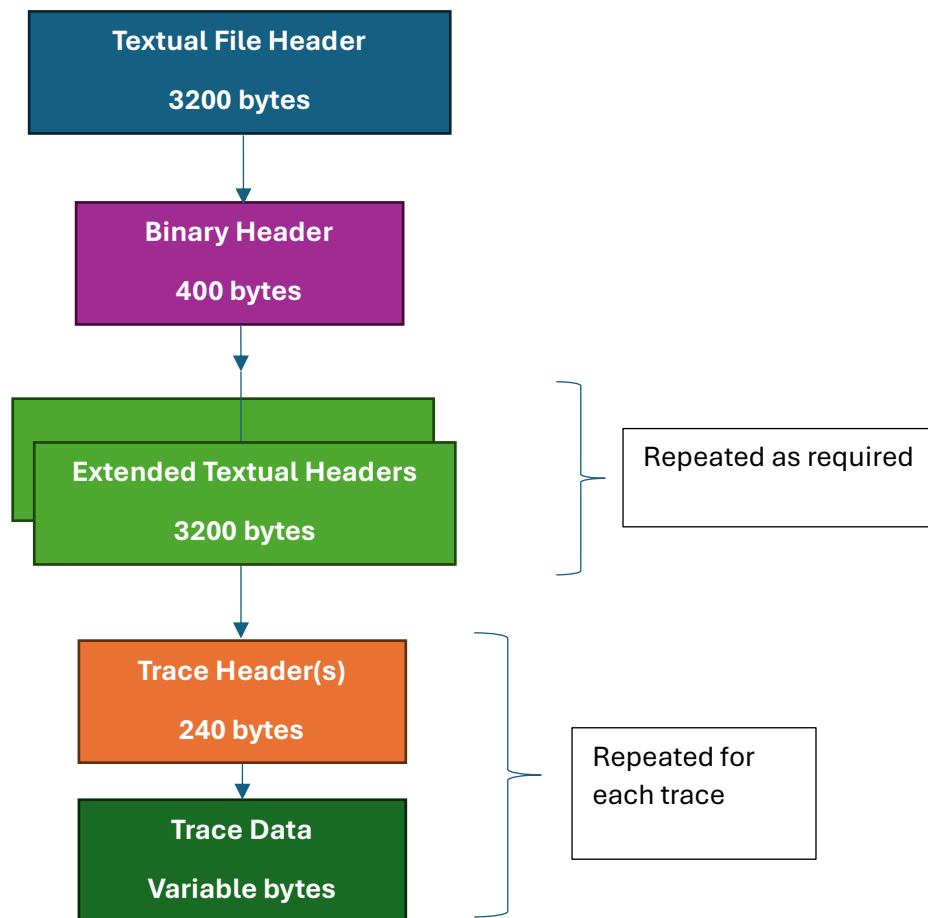
The implementation of the Data Trailer blocks is not covered by this clarification document.

Overview of the Format Revisions

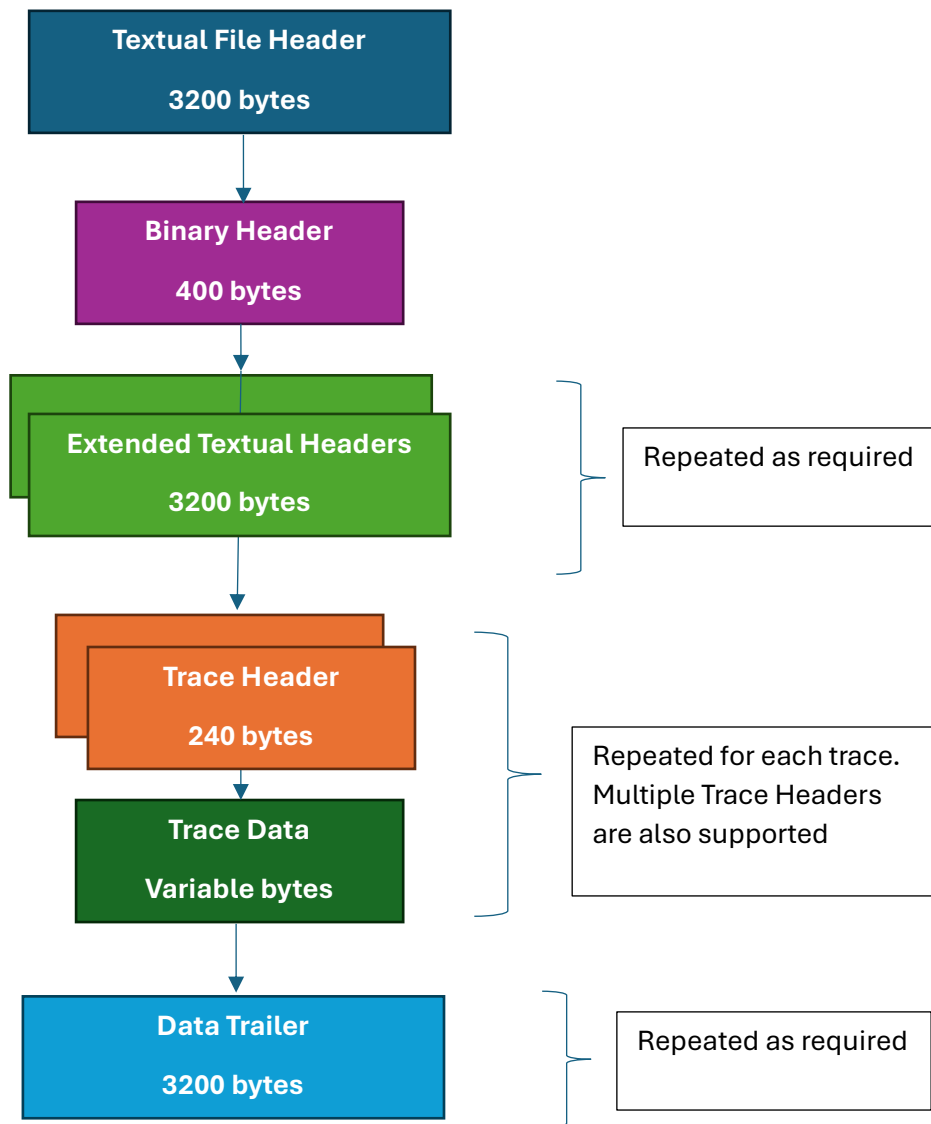
The format of a **Rev 0** SEG-Y file is shown below.



Rev 1 of the SEG-Y format introduced the idea of a repeatable Extended Textual Header block of 3200 bytes which could be inserted between the Binary Header and the Trace Header/Data pairs.



Then **Rev 2.0** of the SEG-Y format introduced the idea of a repeatable Data Trailer block of 3200 bytes which could be appended after the last set of trace data, along with changes to the content of the Extended Textual Headers which were further revised in **Rev 2.1**. The implementation of the Data Trailer blocks is not covered by this clarification document.



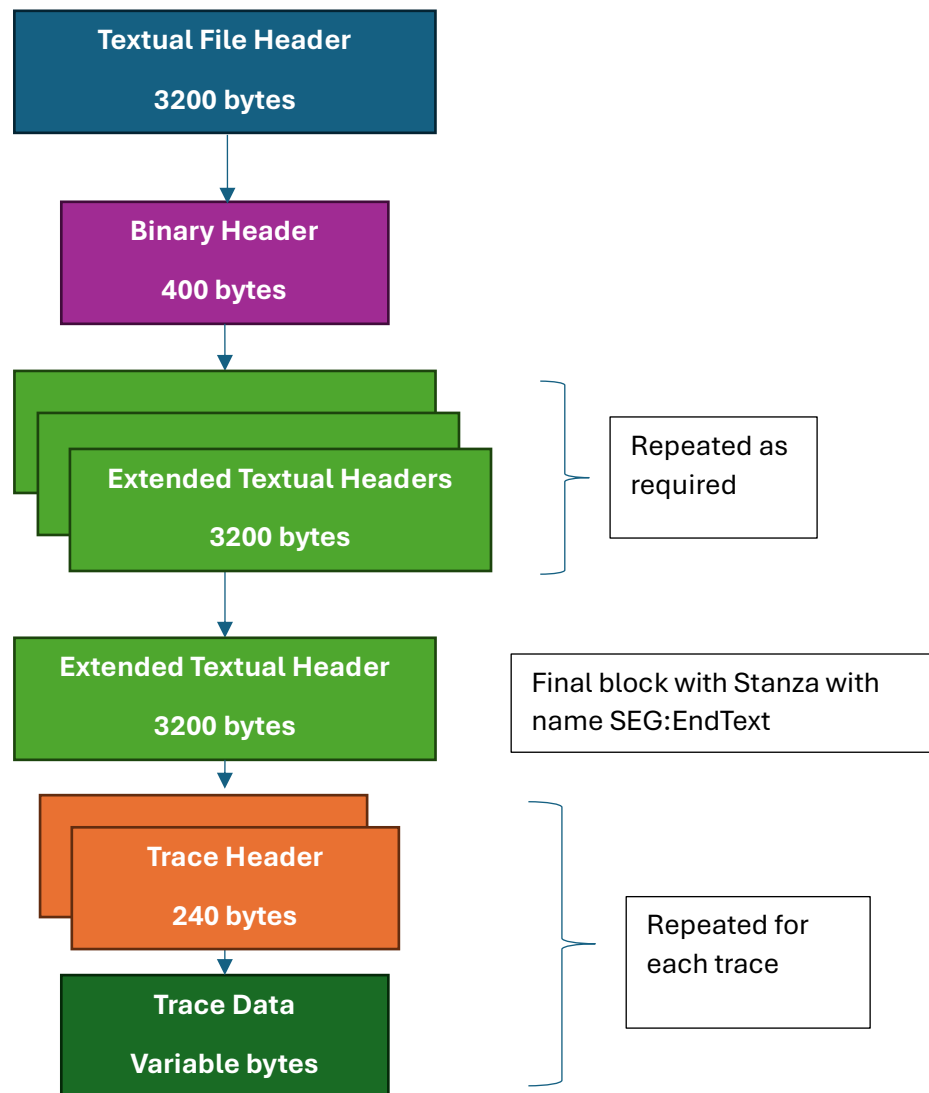
Rev 2.1 also details an XML schema in chapter D.8 for defining the layout of the trace header entries in a SEG-Y file.

Auto-readability and random access: regarding original goals of auto-readability and random access, it is recommended that the Fixed Length Trace Flag binary header 3503-3504 should be set to '1' per the SEG-Y rev 2.1 standard.

Implementation of Extended Textual Header blocks

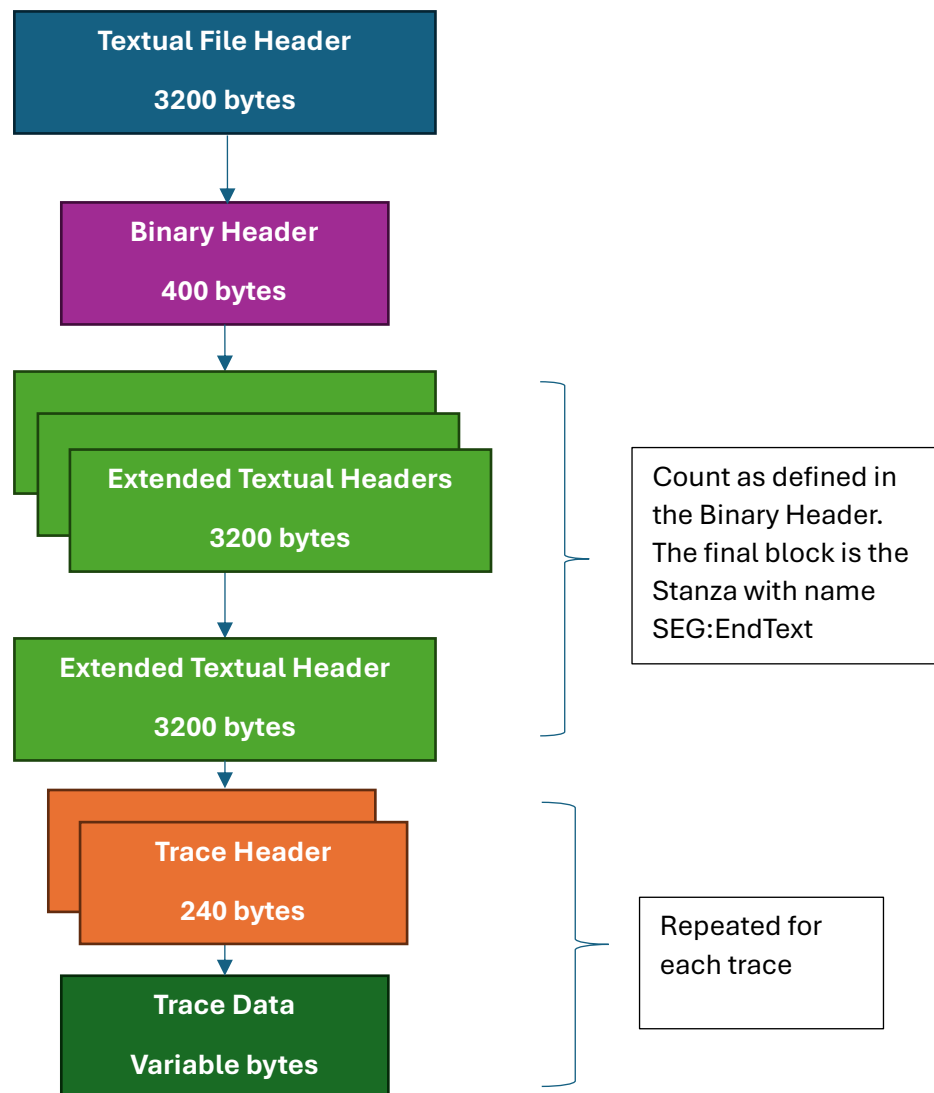
The format of a SEG-Y file with Extended Textual Header blocks depends on the value logged in the binary header value “**Number of 3200-byte, Extended Textual File Header records following the Binary Header**” at bytes 3505-3506.

If this value is -1, the Extended Textual Header blocks are logged as below.



Applications reading the file should read the Textual File Header and Binary Header and then determine if the subsequent block is an Extended Textual Header by checking for the presence of a Stanza Name at the beginning of the block. This, and subsequent Extended Textual Header blocks should be read until the SEG:EndText **Stanza Name** block is read, following which the trace header and data blocks should be read as normal. Note a stanza may span as many 3200-byte blocks as needed, the presence of the next **Stanza Name** or SEG:EndText **Stanza Name** signalling the end of that Extended Textual Header.

If the “**Number of 3200-byte, Extended Textual File Header records**” parameter is defined in the header, the Extended Textual Header blocks are logged as shown below. The last block must still contain the SEG:EndText **Stanza Name**.



If a software implementation wishes to use the “**Number of 3200-byte, Extended Textual File Header records**” parameter to read the file, read the Textual File Header and Binary Header and then determine the number of Extended Textual Header blocks by looking at the value listed in the binary header. The Extended Textual Header blocks should be read, following which the trace header and data blocks should be read as normal.

Since the Extended Textual Headers in rev 2.1 must be created with a final block containing the SEG:EndText stanza, it is also possible to count the number of Extended Textual Header blocks in the file using the method suggested on the previous page, above, for files where the parameter is set to -1.

Once the Textual File Header, Binary Header and Extended Textual Headers have been read from the SEG-Y file, the Trace Headers and Data blocks should be read as normal.

Applications should not rely on the presence of this parameter as it is not mandatory to write it into the binary header, as it may not be known when the binary header is created.

If the value of the “Number of 3200-byte Extended Textual File Header records” in the Binary Header is set to zero in a rev 2.1 file, then there are no Extended Textual Headers, with the file proceeding from the end of the Binary Header directly to the first Trace Header.

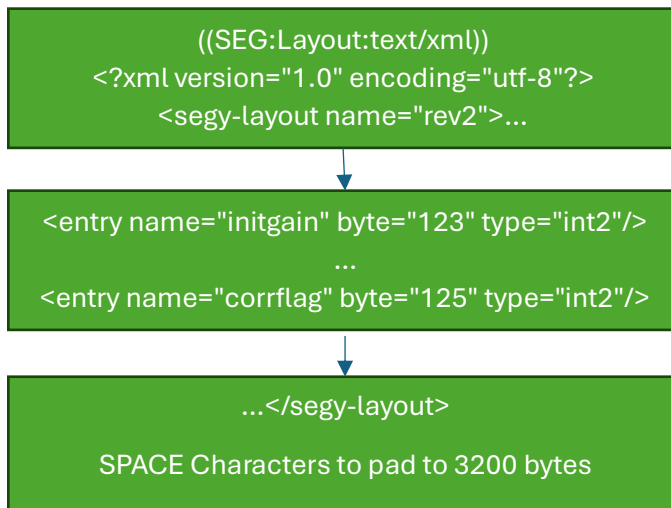
Format of Extended Textual Header Blocks

A document or data stream of any format (e.g., ASCII Text, XML, JSON, PNG, etc.) can be embedded into the Extended Textual Header blocks. Although this extends the blocks to contain non-textual data, we have kept with the “Extended Textual Header” name to be consistent with the original SEG-Y rev 2.1 exchange format definition.

A restriction to the formatting of Extended Textual Headers is made for the Layout XML stanza, which must be written using XML.

When a document or data stream **Stanza** is embedded in the Extended Textual Header blocks, the first block starts with a **Stanza Name** detailing the type, content and optionally the size of the **Stanza**. The **Stanza** data is then embedded into the remaining space within the first and subsequent Extended Textual Header blocks. The **Stanza Name** is only listed in the first block.

Any leftover space in the last Extended Textual Header block should be padded with SPACE characters (0x20 for ASCII or 0x40 for EBCDIC) for text data or NULL characters (0x00) for binary data.



Stanza Name Format

The **Stanza Name** at the top of each **Stanza** should begin with double left parentheses "(" (EBCDIC 0x4D or ASCII 0x28) characters and end with double right parentheses ")" (EBCDIC 0x5D or ASCII 0x29) characters. Any content following the closing parentheses is considered to be part of the **Stanza** data.

The **Stanza Name** consists of two mandatory and two optional text values separated by the character ":" (EBCDIC 0x7A or ASCII 0x3A)

((COMPANY:DATANAME[:CONTENTTYPE][:BYTES]))

For Text data, the **Stanza Name** should list the company or organization name and the data name separated by the character ":" (EBCDIC 0x7A or ASCII 0x3A)

((COMPANY:DATANAME))

The content type of the **Stanza** (as per IANA) may also be included in the **Stanza Name** following the data name, so the **Stanza Name** will read

((COMPANY:DATANAME:CONTENTTYPE))

For binary data, the size of the **Stanza** in bytes must be included in the **Stanza Name** following the content type – the size is the size of the embedded file and must not include the size of the stanza name.

((COMPANY:DATANAME:CONTENTTYPE:BYTES))

Example Stanza Names

- ANY:SOMETEXT
- SEG:LAYOUT:text/xml
- SEG:Catalog:application/json
- OGP:P1/11:text/csv
- TROIKA:IMAGEPNG:image/png:51929

Examples of IANA content-type¹

- text/xml
- text/csv
- application/json
- image/png
- image/bmp
- application/vnd.shp
- application/vnd.shx
- application/vnd.dbf

¹ For a comprehensive list of additional content types, see IANA (Internet Assigned Numbers Authority), 2025, *Media Types*, <https://www.iana.org/assignments/media-types/media-types.xhtml>.

Rules

The following rules apply to the logging of **Stanza** using the Extended Textual Header blocks in a SEG-Y rev 2.1 format file:

1. One of the **Stanzas** stored in the Extended Textual Header blocks should be the Layout XML, logged natively in XML using any encoding (which should be listed). This XML content should not be converted to EBCDIC unnecessarily.
2. It is recommended that all rev 2.1 format files must list the Layout XML **Stanza**
3. All **Stanzas** must be written in their native format and must not be converted unnecessarily during the writing process.
4. Any content following the closing parentheses of the **Stanza Name** is considered to be part of the Stanza data.
5. Different documents in EBCDIC and ASCII may be mixed into a single SEG-Y file, there is no requirement they should be all EBCDIC or all ASCII.
6. Any space in the last block of a **Stanza** should be padded with SPACE characters (for Text files) and NULL characters (for Binary files).
7. The last Extended Textual Header block should be the SEG: EndText block, regardless of the value listed in the Number of Extended Textual Header Blocks parameter in the binary header, (unless this value is set to zero, in which case there are no Extended Textual Header blocks, including the SEG:EndText block). The **Stanza Name** for this must be **((SEG:EndText))**. The rest of this block should be padded with SPACE characters.
8. To be brought into compliance with SEG-Y rev 2.1, legacy SEG-Y data with trace header data in incorrect byte offset locations must contain a Layout XML **Stanza** specifying corresponding trace locations with supporting information.

Example Extended Textual Header for SEG:Layout ASCII Encoding

00000000	28 28 53 45 47 3A 4C 61 79 6F 75 74 3A 74 65 78	((SEG:Layout:tex t/xml))<?xml ver sion="1.0" encod ing="UTF-8" stan dalone="no" ?><s egy-layout><entr y byte="1" name= "linetrc" type="
00000010	74 2F 78 6D 6C 29 29 3C 3F 78 6D 6C 20 76 65 72	
00000020	73 69 6F 6E 3D 22 31 2E 30 22 20 65 6E 63 6F 64	
00000030	69 6E 67 3D 22 55 54 46 2D 38 22 20 73 74 61 6E	
00000040	64 61 6C 6F 6E 65 3D 22 6E 6F 22 20 3F 3E 3C 73	
00000050	65 67 79 2D 6C 61 79 6F 75 74 3E 3C 65 6E 74 72	
00000060	79 20 62 79 74 65 3D 22 31 22 20 6E 61 6D 65 3D	
00000070	22 6C 69 6E 65 74 72 63 22 20 74 79 70 65 3D 22	
00000080	6C 69 6E 65 74 72 63 22 2F 3E 3C 65 6E 74 72 79	
00000090	20 62 79 74 65 3D 22 35 22 20 6E 61 6D 65 3D 22	
000000a0	72 65 65 6C 74 72 63 22 20 74 79 70 65 3D 22 72	linetrc" type="r
000000b0	65 65 6C 74 72 63 22 2F 3E 3C 65 6E 74 72 79 20	eeltrc"/><entry

EBCDIC Encoding

00000000	4D 4D E2 C5 C7 7A D3 81 A8 96 A4 A3 7A A3 85 A7	MM...Z.....Z... .a...]]Lo...@...~..K..@..... ...~.....`..@..... .....~.....@onL.. ...`.....nL..... .@.....~...@.....~@.....~..anL..... @.....~...@.....~..@.....~..anL.....@
00000010	A3 61 A7 94 93 5D 5D 4C 6F A7 94 93 40 A5 85 99	
00000020	A2 89 96 95 7E 7F F1 4B F0 7F 40 85 95 83 96 84	
00000030	89 95 87 7E 7F E4 E3 C6 60 F8 7F 40 A2 A3 81 95	
00000040	84 81 93 96 95 85 7E 7F 95 96 7F 40 6F 6E 4C A2	
00000050	85 87 A8 60 93 81 A8 96 A4 A3 6E 4C 85 95 A3 99	
00000060	A8 40 82 A8 A3 85 7E 7F F1 7F 40 95 81 94 85 7E	
00000070	7F 93 89 95 85 A3 99 83 7F 40 A3 A8 97 85 7E 7F	
00000080	93 89 95 85 A3 99 83 7F 61 6E 4C 85 95 A3 99 A8	
00000090	40 82 A8 A3 85 7E 7F F5 7F 40 95 81 94 85 7E 7F	
000000a0	99 85 85 93 A3 99 83 7F 40 A3 A8 97 85 7E 7F 99	
000000b0	85 85 93 A3 99 83 7F 61 6E 4C 85 95 A3 99 A8 40	

Full details on the Layout XML format are contained in section D-8 of the SEG-Y_r2.1 format description (https://library.seg.org/seg-y_rev2-1)².

² SEG Technical Standards Committee, 2023, *SEG-Y_r2.1: SEG-Y revision 2.1 Data Exchange format*: Society of Exploration Geophysicists, https://library.seg.org/seg-y_rev2-1.

Example Extended Textual Header for SEG:EndText

ASCII Encoding

00000000	28	28	53	45	47	3A	45	6E	64	54	65	78	74	29	29	20	((SEG:EndText))
00000010	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	
00000020	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	
00000030	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	
00000040	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	
00000050	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	
00000060	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	
00000070	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	
00000080	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	
00000090	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	
000000a0	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	
000000b0	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	20	

EBCDIC Encoding

00000000	4D	4D	E2	C5	C7	7A	C5	95	84	E3	85	A7	A3	5D	5D	40	MM...z.....]]@
00000010	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	@
00000020	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	@
00000030	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	@
00000040	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	@
00000050	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	@
00000060	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	@
00000070	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	@
00000080	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	@
00000090	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	@
000000a0	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	@
000000b0	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	40	@

List of Current Stanzas

The following Stanzas are pre-defined:

Company	Data Name	Content Type	Description
SEG	Layout	text/xml	Trace Header Definition as defined in the SEG-Y rev 2.1 Format
SEG	EndText		Terminating Stanza as defined in the SEG-Y rev 2.1 Format
SEG	Catalog	application/json ³	Additional Survey Information
OGP	P1/11	text/csv	IOGP P1/11 Data
OGP	P6/11	text/csv	IOGP P6/11 Data
OGP	P1/90	text/plain	IOGP P1/90 Data
OGP	P6/98	text/plain	IOGP P6/98 Data

The IOGP formats mentioned above can be downloaded from the IOGP website (<https://www.iogp.org/>)⁴.

User-defined Stanzas can be defined as required.

³ For the official registration and details of the application/json media type, see IANA, *Media Type Registration: application/json*, <https://www.iana.org/assignments/media-types/application/json>. Official registration and details of other media types can be found at <https://www.iana.org/assignments/media-types/media-types.xhtml>.

⁴ International Association of Oil & Gas Producers (IOGP), *IOGP*, <https://www.iogp.org/>.